MILESTONE-2 : Data Loading and Data Modeling

DATA LOADING

1. Introduction to Data Source

The Hotel Revenue Analysis report was developed using structured hotel booking and revenue datasets imported into Power BI Desktop. The dataset contains information related to hotel bookings, customer details, room types, stay duration, booking dates, and revenue generated.

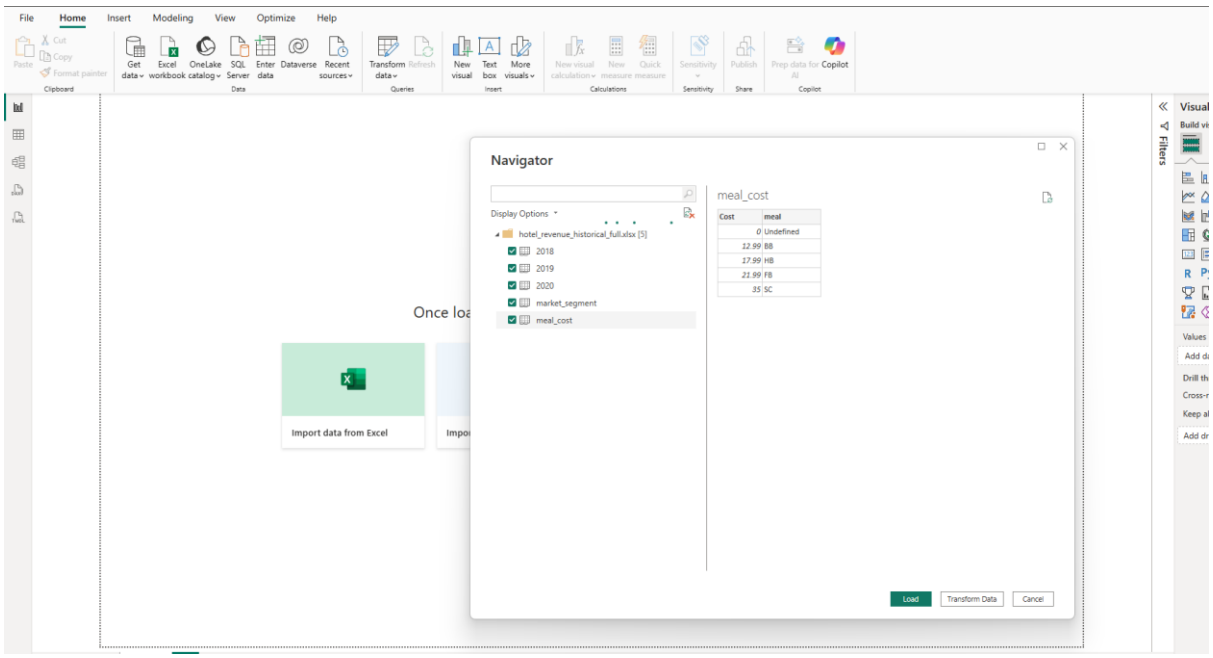
The data was imported into Power BI using the “Get Data” feature from supported sources such as Excel, CSV files, or databases.

2. Data Import Process

The following steps were followed to load the data into Power BI:

- Opened Power BI Desktop.
- Selected the “Get Data” option from the Home tab.
- Chose the appropriate data source.
- Selected the required tables.
- Loaded the data into Power BI.

Multiple related tables were imported to support detailed revenue and booking analysis.



3. Data Cleaning and Transformation

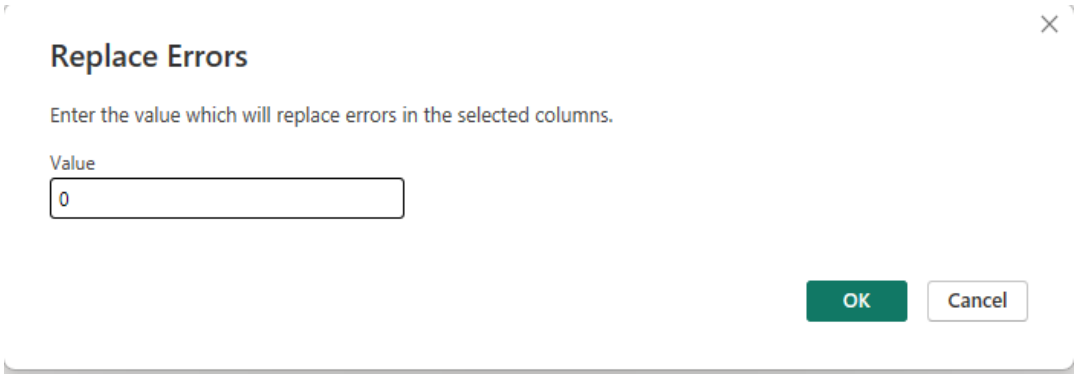
Before loading the data into the report, necessary data transformations were performed using Power Query Editor. These transformations ensured that the dataset was clean, consistent, and ready for analysis.

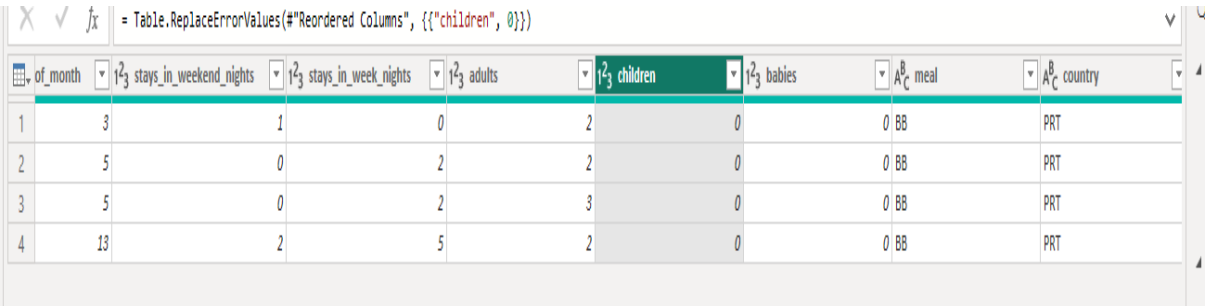
The following data preparation steps were carried out:

- Removed unnecessary or duplicate columns.
- Renamed columns for better clarity and readability.
- Handled missing or null values.
- Corrected data types (Date, Currency, Whole Number, Text).
- Created calculated columns where required (e.g., Total Revenue = Room Rate × Number of Nights).
- Standardized categorical values such as booking status and room types.

After completing these steps, the cleaned data was applied and loaded into the Power BI data model.

Filtering Errors:





Filtered Columns:

	reservation_status
0	Check-Out
0	Check-Out
0	Check-Out
0	Check-Out
1	Check-Out
1	Check-Out
0	Check-Out
1	Check-Out
0	Check-Out
3	Check-Out
1	Check-Out
0	Check-Out
3	Check-Out
0	Check-Out
0	Check-Out
0	Check-Out

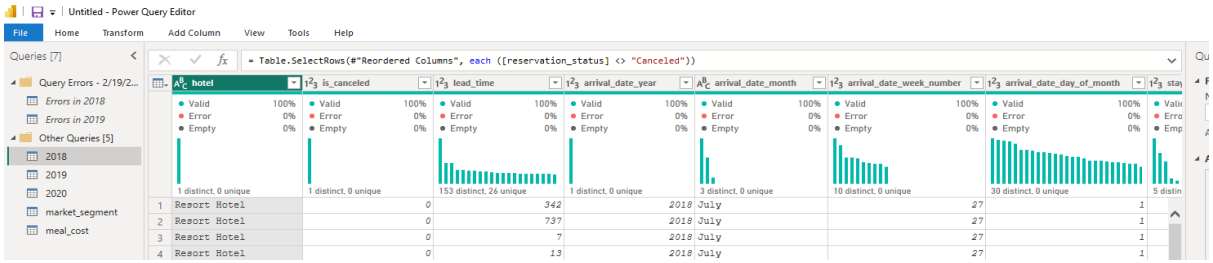
Adding Custom Columns:

	reservation_status	Total Nights	Revenue
0	Check-Out	0	0
0	Check-Out	0	0
0	Check-Out	1	75
0	Check-Out	1	75
1	Check-Out	2	196
1	Check-Out	2	196
0	Check-Out	2	214
1	Check-Out	2	206
0	Check-Out	4	580

4. Data Validation

Basic validation checks were performed to ensure data accuracy:

- Verified that there were no duplicate booking records.
- Confirmed correct date formats.
- Checked revenue calculations for accuracy.
- Ensured consistency in customer and hotel identifiers.



This ensured the reliability of the dataset before proceeding to data modeling.

DATA MODELING

1. Model Structure

The data model was designed using a star schema approach to ensure efficient performance and simplified analysis.

The model consists of:

Fact Table:

- Booking/Revenue Table containing transactional data such as Booking ID, Hotel ID, Customer ID, Room Type, Stay Duration, and Revenue.

Dimension Tables:

- Date Table
- Customer Table
- Hotel Table
- Room Type Table

The fact table is centrally placed and connected to multiple dimension tables using key relationships.

2. Relationships

Relationships were created between tables using One-to-Many (1:*) relationships.

Examples of relationships include:

- Date Table connected to Booking Table using Booking Date.
- Customer Table connected to Booking Table using Customer ID.
- Hotel Table connected to Booking Table using Hotel ID.
- Room Type Table connected to Booking Table using Room Type ID.

These relationships allow accurate filtering and aggregation across reports.

3. Keys and Data Types

Primary keys were defined in dimension tables and linked to corresponding foreign keys in the fact table.

Appropriate data types were assigned:

- Date fields → Date format
- Revenue fields → Currency
- Numerical counts → Whole Number
- Categorical fields → Text

This ensures proper calculations and report performance.

4. DAX Measures

Several DAX measures were created to support hotel revenue analysis. These include:

- Total Revenue
- Total Bookings
- Average Daily Rate (ADR)

These measures allow dynamic and interactive analysis based on filters such as date, hotel, and room type.

5. Model Optimization

To improve performance and efficiency:

- Unnecessary columns were removed.
- Calculated measures were preferred over calculated columns.
- Relationships were optimized to avoid ambiguity.
- Data was structured in a star schema format.
- This ensures faster report performance and better analytical capabilities.

Conclusion

The data loading and data modeling process ensured that the dataset was clean, structured, and logically connected. By implementing a star schema model and creating relevant DAX measures, the Power BI report effectively supports detailed hotel revenue analysis across multiple dimensions such as time, customer type, hotel, and room category.